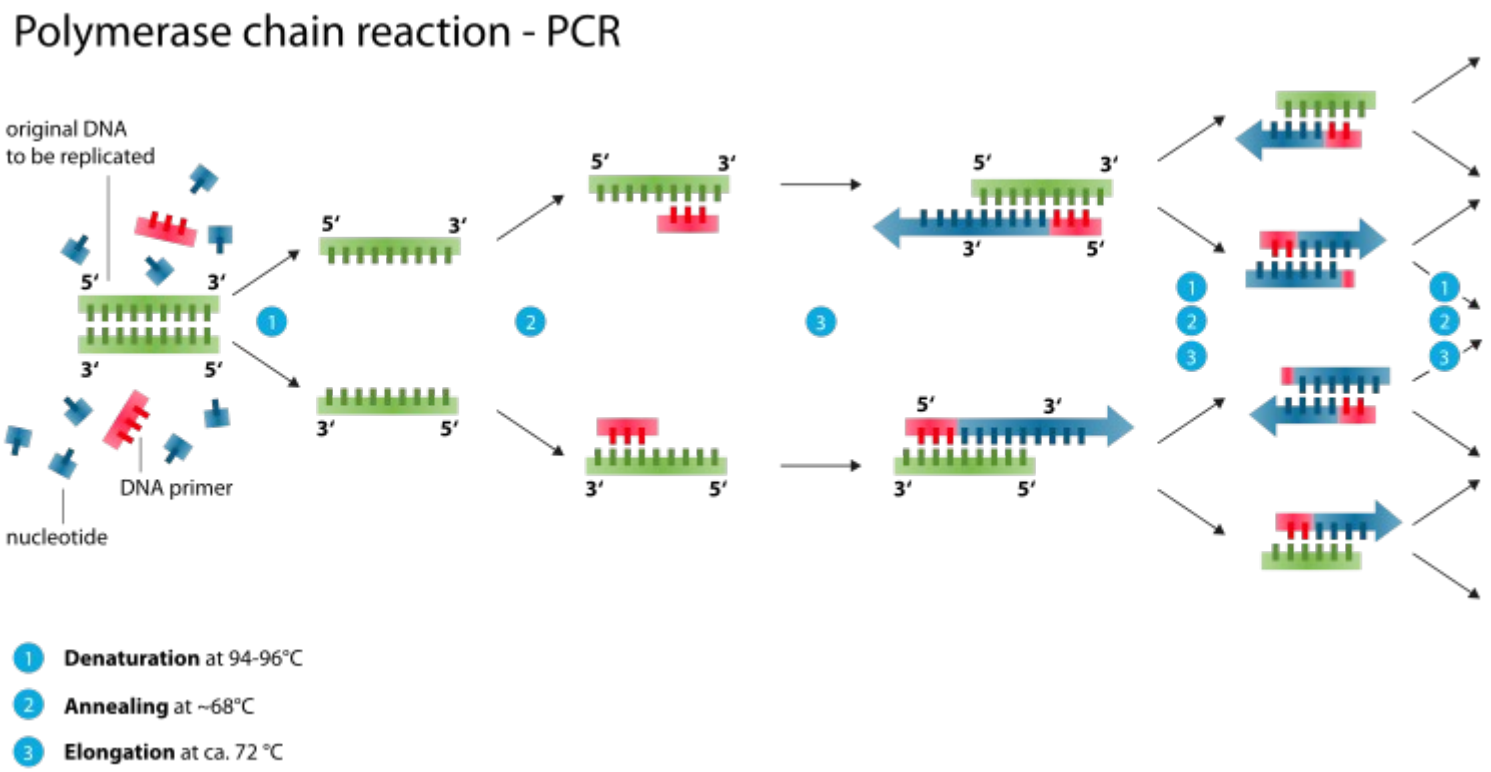
Techniques in Biotechnology



Human Genome

Biotechnology and DNA

Polymerase Chain Reaction (PCR)

Biotechnology

- Use of technology to manipulate organisms for human advantage

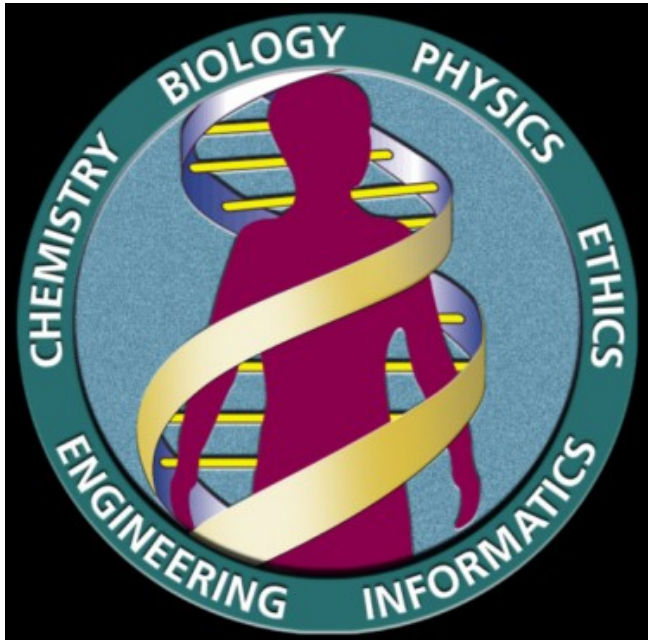
- Can be simple:
  - Using yeast to make bread
  - Cultures for yoghurt and cheese

- Can be complex and modern:
  - Genetic testing
  - Gene manipulation
  - Cell replacement therapies
  - Tissue engineering

Learning Aim: Define Biotechnology and give some examples

The Human Genome Project

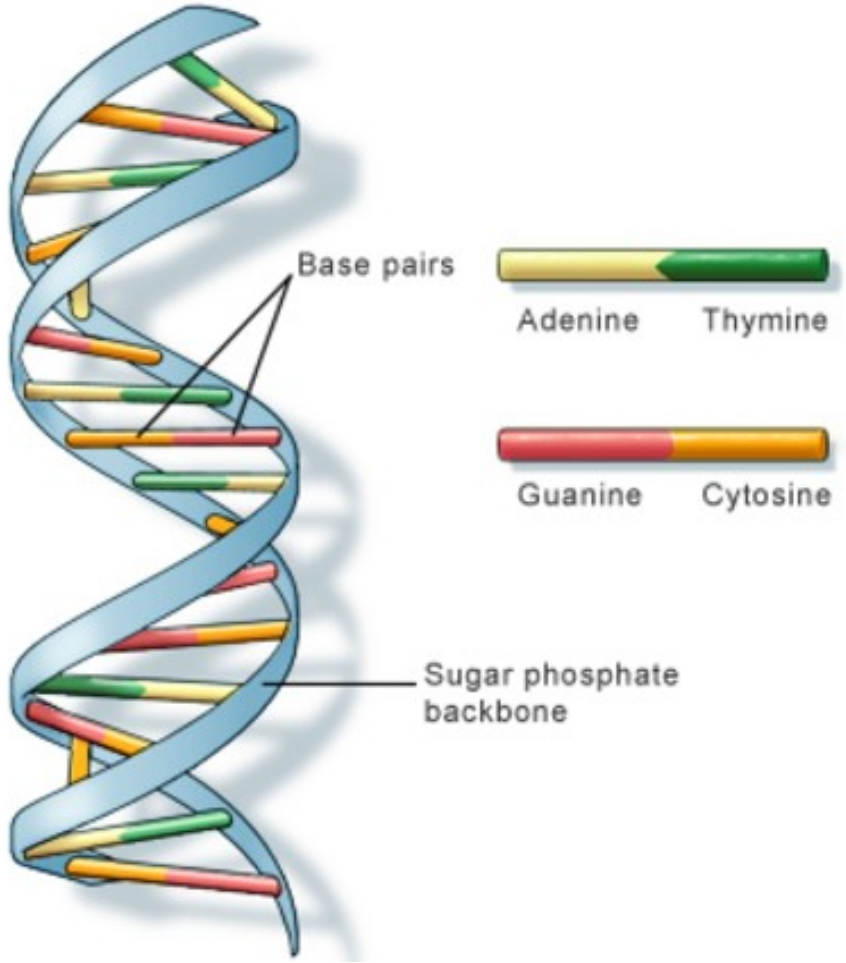
- Process of mapping gene locations on chromosomes
- Project established in 1990
- Complete sequence of nucleotides for the 21000 genes in the human genome
- Project completed in 2003, but data analysis will take much longer
- Potential benefits include:
  - Identification of location of faulty genes on chromosomes
    - For diagnosis
    - For replacement using recombinant DNA technology or tissue engineering
  - Monitoring of gene expression – pathways of genetic disease
  - Genetic testing for disease risk and recommendations to avoid



Learning Aim: Describe the Human Genome Project and outline potential benefits

DNA reminder

- DNA: Deoxyribonucleic acid
  - Found in cells of all organisms
  - Nucleus and mitochondria
  - Sugar and phosphate backbone
  - Nitrogen base “rungs”
  - 4 bases:
    - Adenine (A)
    - Thymine (T)
    - Cytosine (C)
    - Guanine (G)
  - Order of bases codes for protein production



Learning Aim: Understand the structure of DNA

Polymerase Chain Reaction (PCR)

- Generates millions of copies of DNA from a small sample

- Shortens the time taken to sequence DNA

    - Each gene of interest can be amplified so easier to work with
    - Quicker detection of disease mutations eg Cystic fibrosis
    - Can detect viral diseases early in infection
    - Can get a large amount of DNA from a very small biological sample eg fossils, single drop of blood.

Learning Aim: Describe the purpose of PCR

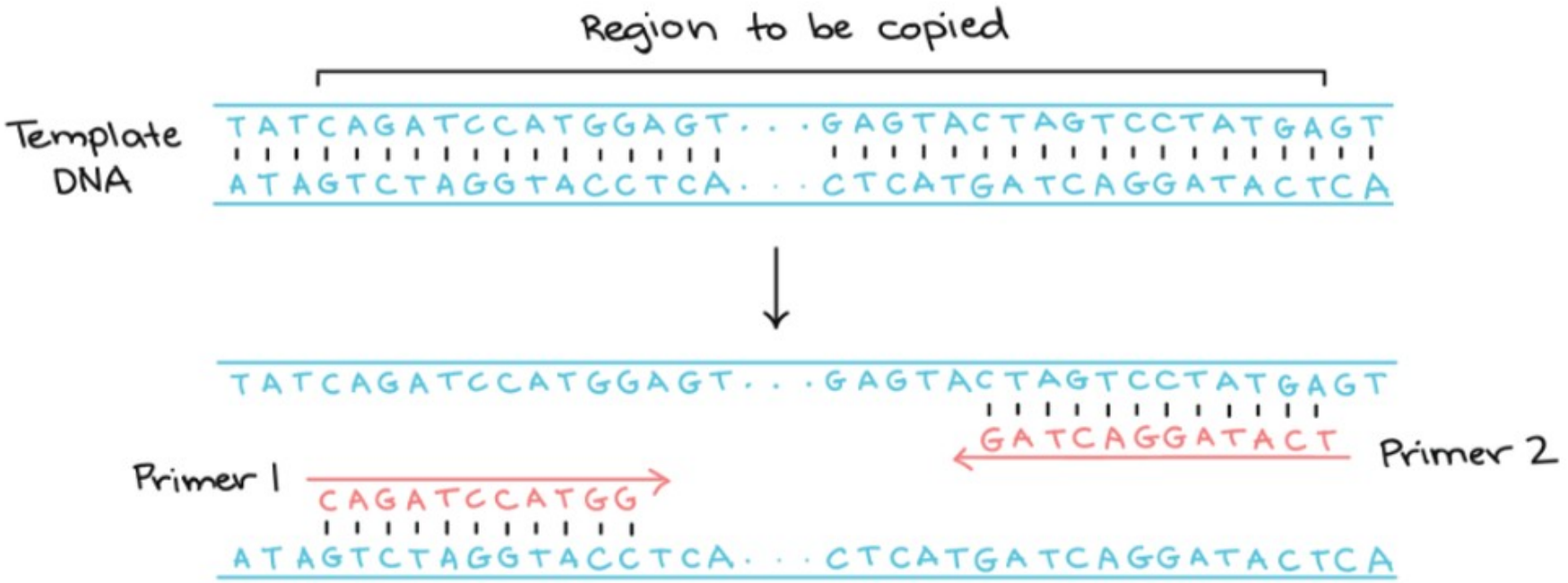
Development of PCR

- Developed by Kary Mullis, who was awarded a Nobel Prize for Chemistry, in 1993
- Human DNA polymerase assists replication BUT is denatured during heating to separate strands of DNA.
- Mullis used heat-stable DNA polymerase from the bacterium *Thermus aquaticus*, which inhabits very hot aquatic environments.
- This heat stable DNA polymerase is called *Taq polymerase*, after the bacterium.
  - *Taq* polymerase enzyme is most active at around 70°C (a temp at which human DNA polymerase would denature). This means it stays stable through the cycles of temperature change in PCR.

Learning Aim: Outline how the PCR technique was developed.

PCR: Primers

- Short pieces of single strand DNA, around 20 nucleotides in length.
- Used as an attachment point for *Taq* polymerase to begin to add new nucleotides.
- Two primers used in PCR, one opposite ends of each strand of the DNA sample of interest.



Learning Aim: Outline the ingredients of PCR and their functions

PCR Ingredients and Steps

- PCR is a technique to make many copies of a DNA sample.
- Key ingredients:
  - The DNA sample of interest (as a template)
  - Thermostable DNA polymerase (*Taq* polymerase)
  - DNA Primers specifically for the DNA sample of interest
  - Nucleotides to attach to the template strand
- In PCR, the reaction is repeatedly cycled through a series of temperature changes:

1: Denaturation (96°C): Heat separates the DNA strands to provide single stranded templates. Note: the *Taq* polymerase is heat stable so it doesn't denature and can be used again in the next cycle.

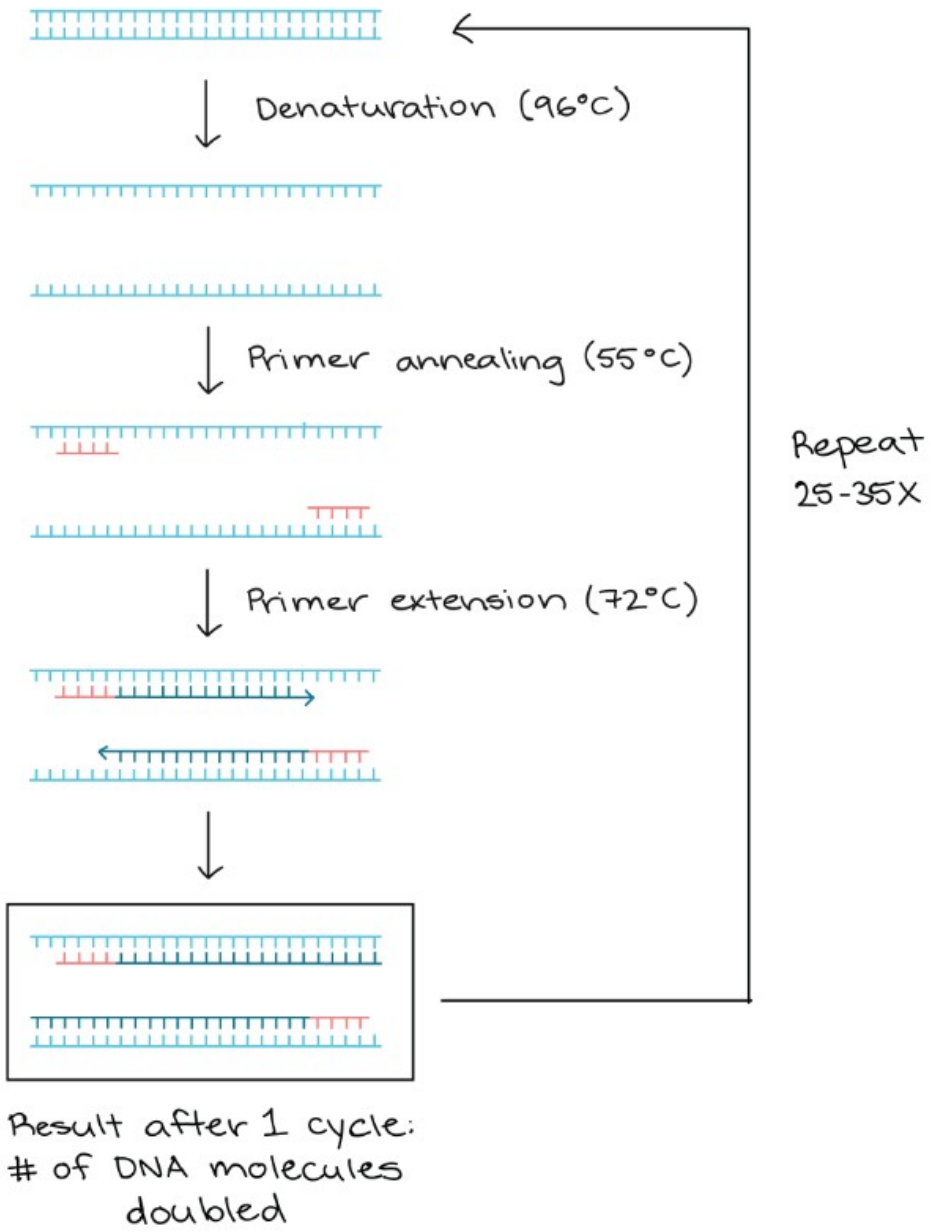
2: Annealing (55-65°C): The reaction is cooled so the primers can anneal (bind) to the template DNA to create a starting point for nucleotides to be added.

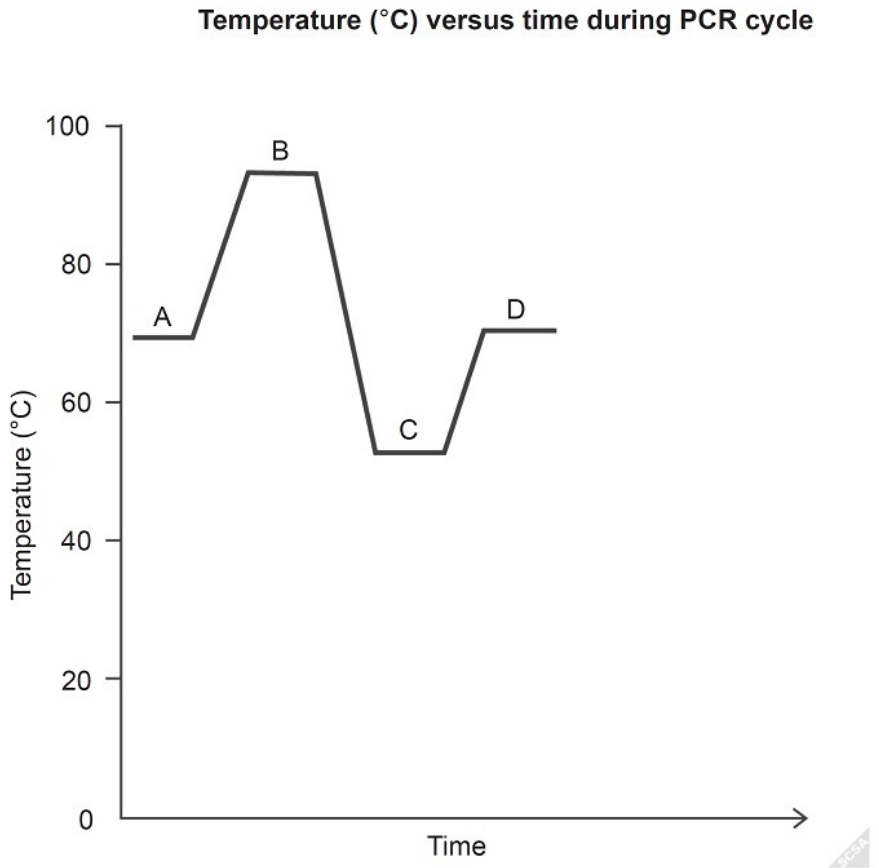
3: Extension (72°C): The temperature is raised so that Taq polymerase can add new nucleotides to the primer, extending the copy of the template strand.

Learning Aim: Outline the ingredients of PCR and their functions

Learning Aim: Describe the PCR cycle in detail

PCR Cycle





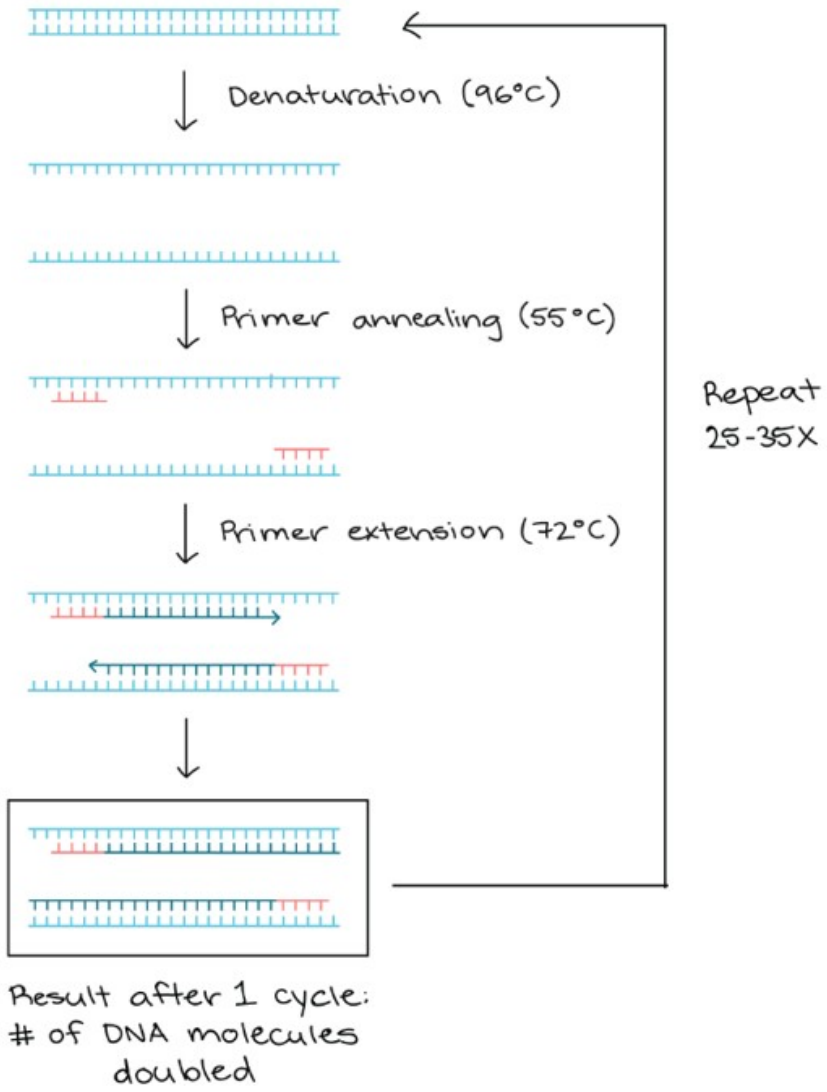
A, D: extension/strand synthesis

B: Denaturation

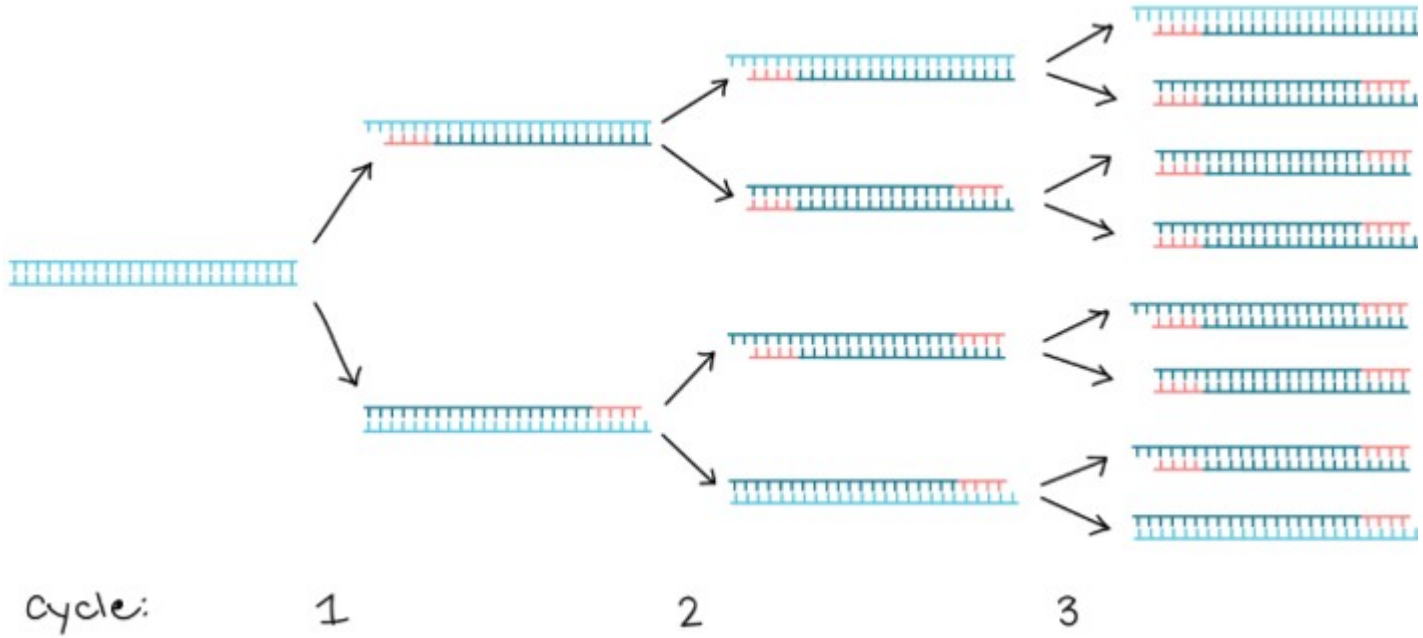
C: Primer Annealing

Learning Aim: Describe the PCR cycle in detail

PCR Cycling



- Cycle repeats 25 -35x on average. The number of DNA molecules doubles each time. Process takes 2-4 hours.
- Can go from a very small sample to billions of copies in this time.
- This is because:
  - The *Taq* polymerase remains functional for the entire process, through the many temperature changes
  - Each cycle, all of the DNA available is denatured, so each cycle the template strands available doubles.



Learning Aim: Describe the PCR cycle in detail

https://www.youtube.com/watch?v=iQsu3Kz9NYo

Applications of PCR

- Amplifies DNA samples for:
  - Forensic identification using gel electrophoresis
  - Cloning to use in recombinant DNA technology
  - Sequencing of DNA to identify genetic diseases
  - To test for the presence of bacterial or viral DNA/RNA to diagnose disease (eg COVID!)

Learning Aim: List applications of PCR